THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Ethanol reacts with sodium to give: 1

(a) oxygen **(b)** ethene **(c)** carbon dioxide **(d)** hydrogen

2. Vinegar is a dilute solution of: 1

(a) ethanal **(b)** methanoic acid **(c)** ethanoic acid **(d)** ethanol

3. Addition reaction is characteristic of: 1

(a) soaps **(b)** alkanes

(c) saturated hydrocarbons **(d)** unsaturated hydrocarbons

4. Which compound is a ketone? 1

(a) CH_3OH **(b)** CH_3CHO **(c)** CH_3COOH **(d)** CH_3COCH_3

5. The product obtained by heating ethanol with excess conc. H_2SO_4 at 443 K is: 1

(a) ethanal **(b)** ethyne **(c)** ethene **(d)** ethane

6. Substitution reaction is characteristic of: 1

(a) saturated hydrocarbons **(b)** unsaturated hydrocarbons

(c) alkynes **(d)** alkenes

7. Which of the following is a homologous series? 1

(a) CH_3OH, CH_3Cl, CH_4 **(b)** C_2H_4, C_3H_8, C_4H_6 **(c)** CH_4, C_2H_4, C_2H_2 **(d)** CH_4, C_2H_6, C_3H_8

8. Which of these hydrocarbons burns with a sooty flame? 1

(a) C_2H_6 **(b)** C_2H_2 **(c)** CH_4 **(d)** C_3H_8

9. Ethanol on oxidation with alkaline $KMnO_4$ gives: 1

(a) ethanal only **(b)** ethene **(c)** ethane **(d)** ethanoic acid

10. The general formula for alkynes is: 1

(a) $CnHn$ **(b)** CnH_2n **(c)** $CnH_2n_2^+$ **(d)** CnH_2n-2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does soap form scum in hard water? 2

12. What are isomers? Give an example. 2

13. Name the products of complete combustion of ethane and write the equation. 2

14. Why are detergents better than soaps in hard water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the cleansing action of soap with micelle formation. 3

16. What is hydrogenation? Where is it used? 3

17. What is a substitution reaction? Give the equation of methane with chlorine. 3

OR

What are the products when ethanol reacts with sodium? Write the equation. 3

18. Why is carbon dioxide produced by burning of ethanol used as clean fuel in some countries? Write the combustion equation. 3

OR

How is ethanoic acid prepared from ethanol and what is glacial acetic acid? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the properties of ethanol: physical, reaction with sodium, dehydration, and uses. 5

OR

Describe allotropes of carbon: diamond, graphite and fullerene with structure and properties. 5

20. Describe the properties of ethanoic acid: reactions with base, carbonate and alcohol, and uses. 5

OR

Why do saturated and unsaturated hydrocarbons burn differently? Describe with example, plus the test to distinguish them using bromine water. 5

END OF QUESTION PAPER — Total: 40 Marks

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